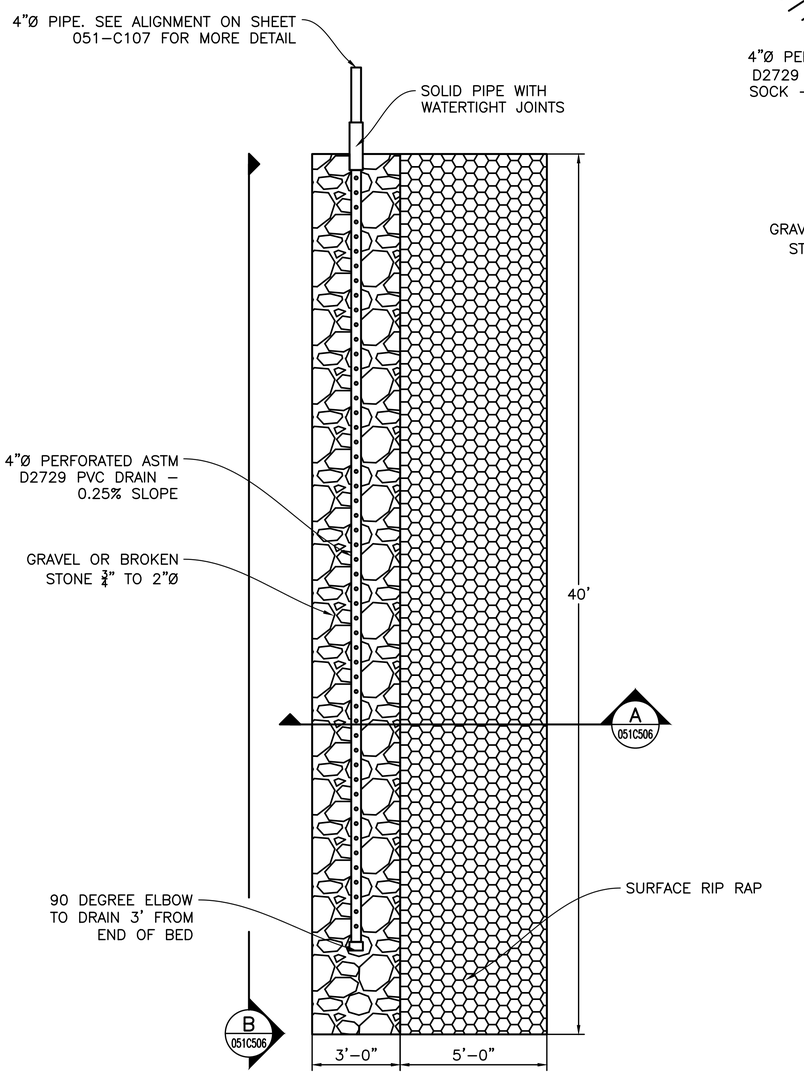
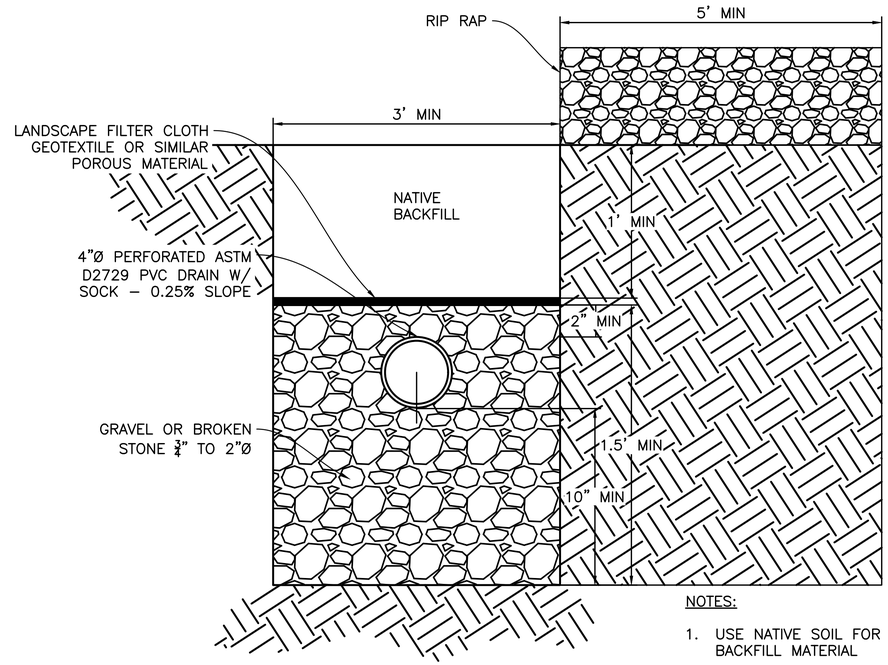


1 051C506 ROARING SPRINGS DRAIN FIELD DETAIL

NOT TO SCALE

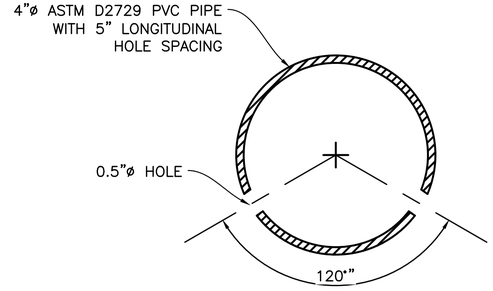


NOTES:

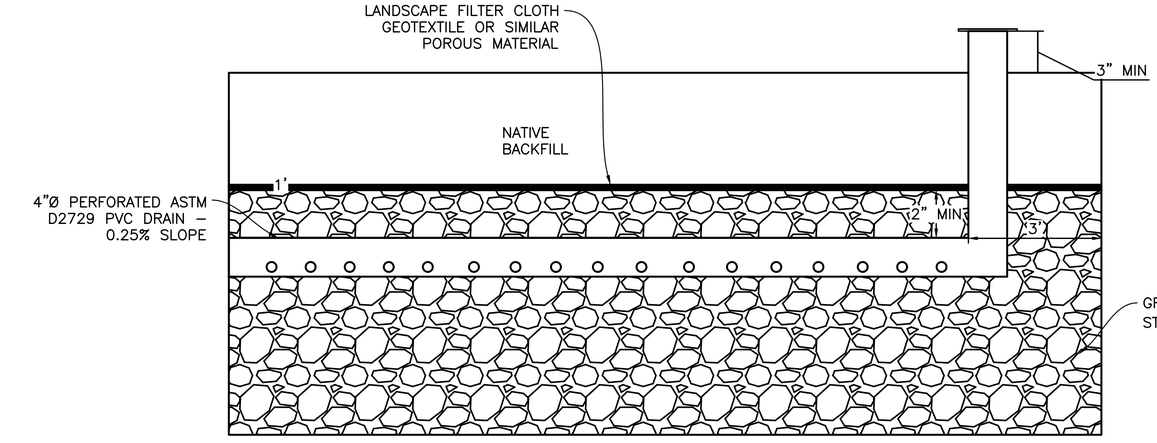
1. USE NATIVE SOIL FOR BACKFILL MATERIAL
2. MAINTAIN 3–FT CLEARANCE FROM CRIB WALL

A 051C506 SECTION

NOT TO SCALE



DRAIN PIPE HOLE PATTERN



B 051C506 SECTION

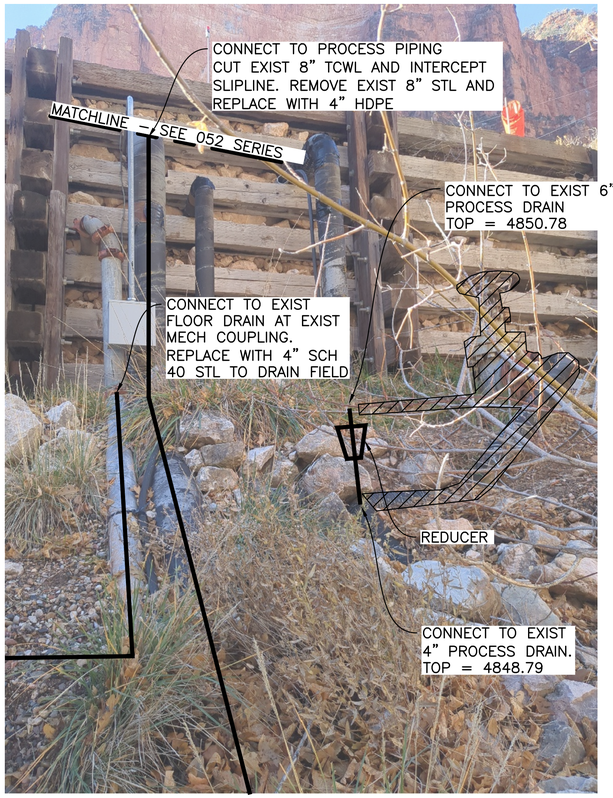
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD T. MCCALLAN TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	051 C506	SITEWORK CIVIL DRAIN FIELD DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 26 OF 113

10/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\CAD_HDR\DRAWINGS\CIVIL\051-C506.dwg



1 / 051C503 RSPH DECK PIPING CONNECTIONS

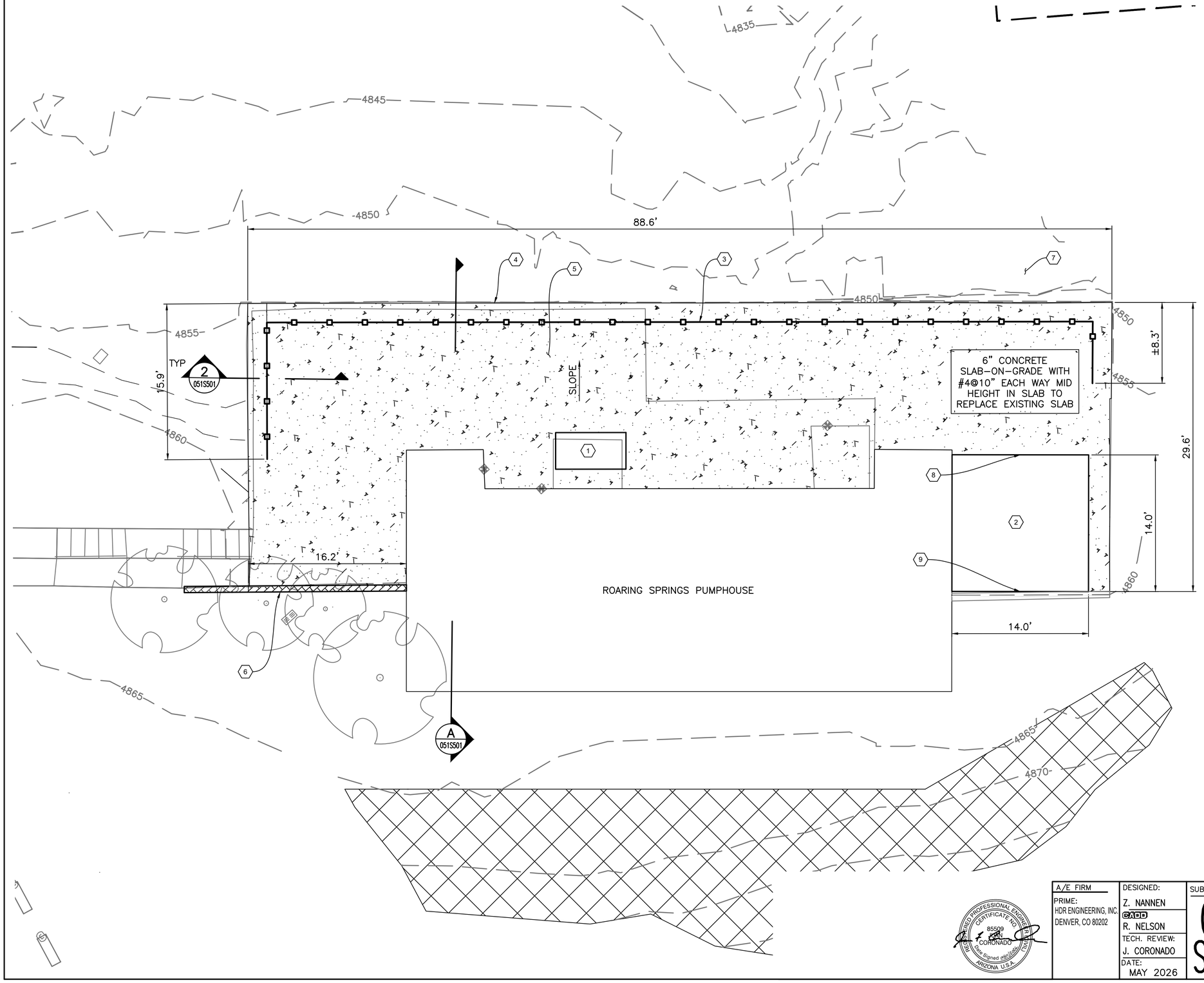
NOT TO SCALE

10/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\051-C901.dwg

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 051 C901	TITLE OF SHEET SITEWORK PHOTO DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 27 OF 113



KEYNOTES (#)

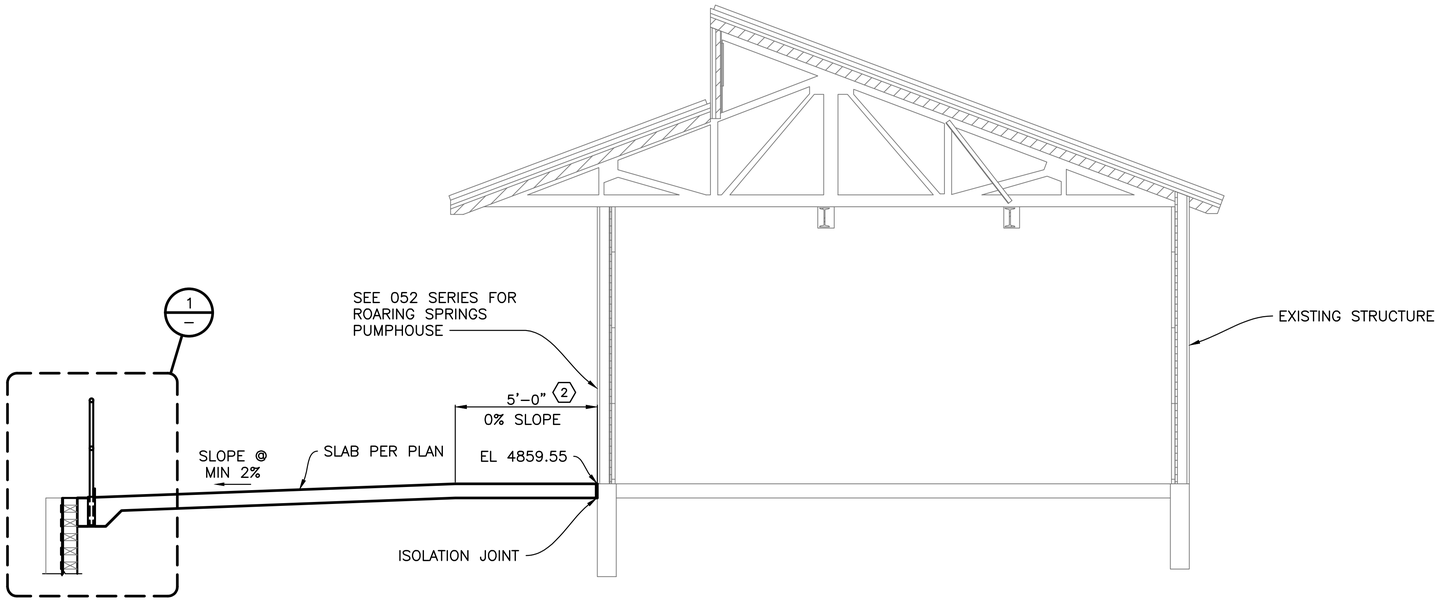
1. EQUIPMENT PAD PER STD DETAIL 2/003S501. COORDINATE LOCATION AND DIMENSIONS WITH MECHANICAL. TOC ELEV = 4859.55.
2. 12" THICK REINFORCED CONCRETE PAD, PER 052S101. EXCAVATE AREA BELOW TANK TO SUITABLE OR NATIVE MATERIAL AS DETERMINED BY CONTRACTING OFFICER'S REPRESENTATIVE. STORAGE TANK SLAB PER SHEET 052S101.
3. MODULAR ALUMINUM REMOVABLE GUARDRAIL FASTENED TO NEW CONCRETE SLAB PER DETAIL 1/051S501 WHERE NEW FINISHED SLAB ELEVATION EXCEEDS 2'-6" ABOVE LOWER GRADE.
4. ENCLOSE THE EXISTING CRIB WALL WITH PRESSURE TREATED WOOD SLAT FENCING. SPACE THE SLATS WITH 1/2 INCH GAPS. EXTEND GEOTEXTILE TO INSIDE FACE OF WOOD SLAT FENCE.
5. PRIOR TO PLACEMENT OF NEW CONCRETE SLAB, REPLACE ANY VOID AREAS BELOW SLAB WITH PROCESSED, MOISTURE CONDITIONED, AND COMPACTED ONSITE MATERIAL LAID ATOP GEOTEXTILE FABRIC. SEE PLAN FOR SLAB REQUIREMENTS.
6. PRECAST MODULAR BLOCK RETAINING WALL PER SPEC 32 32 16. SEE CIVIL.
7. PLACE NATURAL GRAVEL AND STONE AT AREA OF REMOVED VEGETATION IN FRONT OF WALL FOR FUTURE INSPECTION AND MAINTENANCE.
8. ISOLATION JOINT BETWEEN TANK PAD AND SLAB.
9. PROVIDE BOND BREAKER BETWEEN CONCRETE PAD AND EXISTING TIMBER WALL.

5 0 5 10 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JOSEPH CORONADO Digitally signed electronically ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN CADD R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 051 S101	TITLE OF SHEET SITEWORK STRUCTURAL/ GEOTECHNICAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK
			DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 103

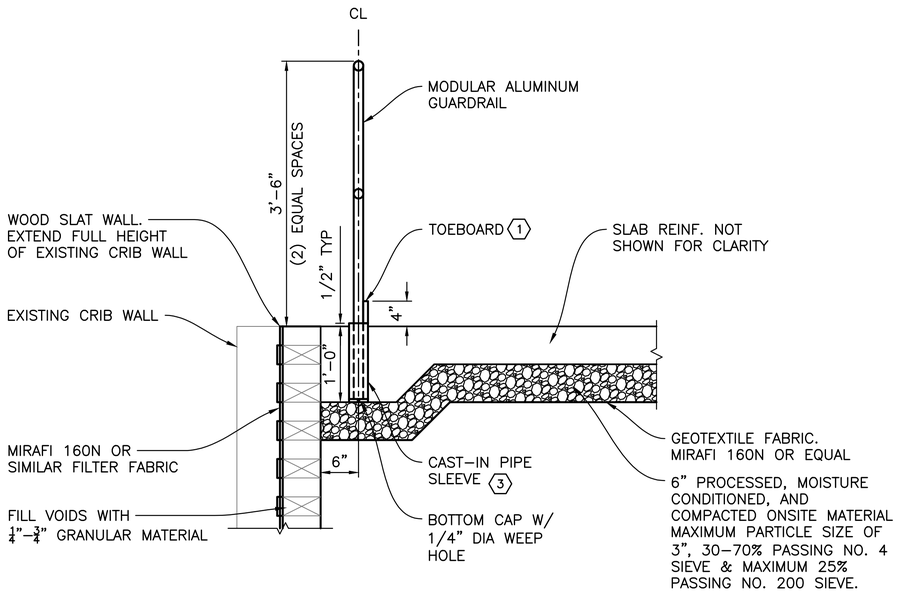
4/27/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\STRUC\051-S101.dwg



A 051S501 **SECTION**
SCALE (A)

KEYNOTES (#)

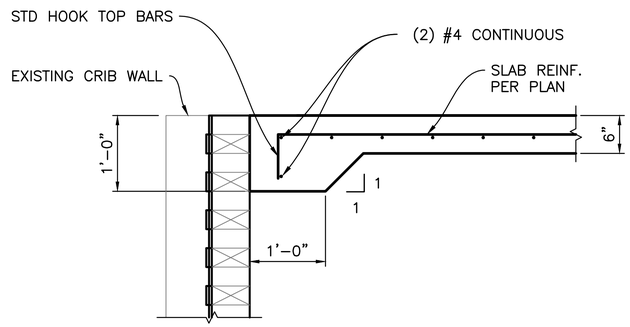
1. TOEBOARD MUST STAY ATTACHED TO MODULAR GUARDRAIL PANELS DURING REMOVAL AND IS ALSO SEPARATELY REMOVABLE FROM PANELS.
2. HOLD 0% SLOPE AT FRONT OF BUILDING BETWEEN WALL PROJECTIONS ONLY AND BELOW ROOF LINE.
3. PROVIDE ISOLATION BETWEEN SLAB REBAR AND GUARDRAIL SLEEVES FOR DISSIMILAR METALS PER SPEC 09 96 00.



1 051S501 **REMOVABLE GUARDRAIL DETAIL**
SCALE (B)

NOTE:

GUARDRAIL NOT SHOWN FOR CLARITY



2 051S501 **SLAB EDGE DETAIL**
SCALE (B)

SCALE (B) 1 0 1 2 1"=1'
SCALE OF FEET

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JOSE CORONADO Date Signed 4/6/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 051 S501	TITLE OF SHEET SITEWORK STRUCTURAL DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 113

4/27/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\STRUC\051-S501.dwg